DI31002: **Information Security**

Assignment Part 1: Applied Cryptography

Deadline for Submission: **23:59 (11.59pm) on** Friday 6**th** November **202**6

Hand in Method: One team member to submit a single **ZIP** file named after the team to the assignment document on My Dundee.

Date Feedback will be Received by: This will be received within the University's 3-week policy which for this assignment is Friday the 2 7 th November 2026.

Penalty for Late Submission: One grade point per day late (meaning if a submission is one day late and marked as a C2 it will receive a C3 grade). A day is defined as each 24-hour period following the submission deadline, including weekends and holidays. Assignments submitted more than 5 days after the agreed deadline will receive a zero mark (LS).

Percentage of Module: Part 1 of the Assignment is worth 50% of the assignment mark (25% of the module).

Assignment Part 1: Applied Cryptography - Marking Scheme

A grade	B grade	C grade	D grade	Fail	Mark
The authentication in digest_auth.py and client.py works correctly, and the files satisfy all the conditions 1a-d, and 2e-g. *10 to 7 marks*	The authentication in digest_auth.py and client.py works, and the files satisfy three of the conditions 1a-d and two of 2e-g. *6 marks*	The authentication in digest_auth.py and client.py works, and the files satisfy two of the conditions 1a-d, and one of 2e-g. *5 marks*	The authentication in digest_auth.py and client.py works and the files satisfy one of the conditions 1a-d and one of 2e-g. *4 marks*	The files digest_auth.py and client.py are not submitted, or do not satisfy any of the conditions 1.a-d and 2.e-g. *3 to 0 marks*	
Comments:					
The authentication in digest_auth_subverted.py and client_subverted.py works correctly, and the files satisfy all conditions 3h-k and 4l-m. *10 to 7 marks*	The authentication in digest_auth_subverted.py and client_subverted.py works, and the files satisfy three of the conditions 3h-k and two of 4l-m. *6 marks*	The authentication in digest_auth_subverted.py and client_subverted.py works, and the files satisfy two of the conditions 3h-k and one of 4l-m. *5 marks*	The authentication in digest_auth_subverted.py and client_subverted.py works and the files satisfy one of the conditions 3h-k and one of 4l-m. *4 marks*	The files digest_auth_subverted.py and client_subverted.py are not submitted, or do not satisfy any of the conditions 3h-k and 4l-m. *3 to 0 marks*	
Comments:					
The vulnerability is located clearly, explained in detail, and its covertness justified. Research was done and alternatives considered. *10 to 7 marks*	The vulnerability is adequately located and explained, and its covertness justified. *6 marks*	One of the elements in the report is not adequately covered. *5 marks*	Two of the elements in the report are inadequately covered. *4 marks*	Inadequate report or no report submitted. *3 to 0 marks*	
Comments:					
All files are submitted in the required format and structure. The report is no longer than 1 page. *10 to 7 marks*	Minor issues in one of the submitted files - e.g., file names, missing team members, etc. *6 marks*	One or more files inadequately submitted or minor issues with two or more files. Extra files submitted. *5 marks*	One file missing. *4 marks*	Two or more files missing. *3 to 0 marks*	
Comments:					
Marks awarded by ranking submissions from shortest to longest in number of "(", ":" and "," used in subverted source code. *10 to 7 marks*	Code is obfuscated as, e.g., HEX strings in order to minimize number of "(", ":" and ",". *6 marks*	Sum of "(", ":" and "," characters in digest_auth_subverted.py is greater than 65. *5 marks*	Sum of "(", ":" and "," characters in digest_auth_subverted.py is greater than 100. *4 marks*	Inadequate subverted authentication submitted (e.g., implemented without hash function.) *3 to 0 marks*	
Comments:					